

13.2 部分习题参考简答

$$1. (1) \int_0^1 \left(\int_{2\sqrt{1-x}}^{\sqrt{4-x^2}} f(x, y) dy \right) dx + \int_1^2 \left(\int_0^{\sqrt{4-x^2}} f(x, y) dy \right) dx = \int_0^2 \left(\int_{1-\frac{y^2}{4}}^{\sqrt{4-y^2}} f(x, y) dx \right) dy;$$

$$(2) \int_1^2 \left(\int_{2-x}^{\sqrt{2x-x^2}} f(x, y) dy \right) dx = \int_0^1 \left(\int_{2-y}^{1+\sqrt{1-y^2}} f(x, y) dx \right) dy;$$

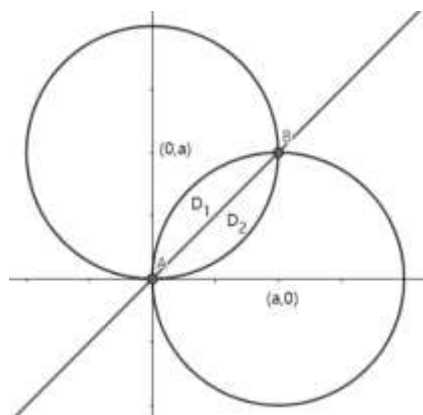
$$(3) \int_0^1 \left(\int_0^{2x^2} f(x, y) dy \right) dx + \int_1^3 \left(\int_0^{3-x} f(x, y) dy \right) dx = \int_0^2 \left(\int_{\frac{y}{2}}^{\frac{3-y}{2}} f(x, y) dx \right) dy;$$

$$(4) \int_{-6}^2 \left(\int_{\frac{x^2-4}{4}}^{2-x} f(x, y) dy \right) dx = \int_{-1}^0 \left(\int_{-2\sqrt{1+y}}^{2\sqrt{1+y}} f(x, y) dx \right) dy + \int_0^8 \left(\int_{-2\sqrt{1+y}}^{2-y} f(x, y) dx \right) dy.$$

$$2. (1) \int_0^1 \left(\int_{y-1}^{1-y} f(x, y) dx \right) dy \text{ 或 } \int_{-1}^0 \left(\int_0^{1+x} f(x, y) dy \right) dx + \int_0^1 \left(\int_0^{1-x} f(x, y) dy \right) dx;$$

$$(2) \int_{-1}^2 \left(\int_{y^2}^{y+2} f(x, y) dx \right) dy \text{ 或 } \int_0^1 \left(\int_{-\sqrt{x}}^{\sqrt{x}} f(x, y) dy \right) dx + \int_1^4 \left(\int_{x-2}^{\sqrt{x}} f(x, y) dy \right) dx.$$

3. 积分区域如图所示



$$\text{令 } \begin{cases} x = \rho \cos \theta \\ y = \rho \sin \theta \end{cases},$$

$$\text{则 } D_1 = \{(\rho, \theta) \mid 0 \leq \rho \leq 2a \sin \theta, 0 \leq \theta \leq \frac{\pi}{4}\},$$

$$D_2 = \{(\rho, \theta) \mid 0 \leq \rho \leq 2a \cos \theta, \frac{\pi}{4} \leq \theta \leq \frac{\pi}{2}\}$$

$$\text{注意到 } \left| \det \frac{\partial(x, y)}{\partial(\rho, \theta)} \right| = \rho, \text{ 所以}$$

$$\iint_D f(x, y) dx dy = \int_0^{\frac{\pi}{4}} d\theta \int_0^{2a \sin \theta} f(\rho \cos \theta, \rho \sin \theta) \rho d\rho + \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} d\theta \int_0^{2a \cos \theta} f(\rho \cos \theta, \rho \sin \theta) \rho d\rho.$$

$$5. (1) \pi. \quad (2) \frac{\pi}{12}. \quad (3) \frac{1}{2} R^4. \quad (4) \frac{1}{12}. \quad (5) \frac{\pi}{2}.$$

$$6. (1) -2. \quad (2) \frac{45}{16} + \ln 2. \quad (3) \frac{2}{15}. \quad (4) \frac{2}{3}. \quad (5) -\frac{175}{54}.$$

$$7. (1) \frac{3\pi}{4}. \quad (2) \frac{a^2}{6}. \quad (3) 4a^2. \quad (4) \frac{5}{8} \pi a^2.$$

$$8. (1) \text{ 位于 } Oxy \text{ 平面以下部分的体积 } \frac{5}{6} - \frac{\pi}{4}; \text{ 位于 } Oxy \text{ 平面以上部分的体积 } \frac{5}{6} + \frac{3\pi}{4}.$$

(2) 6π . (3) $\frac{13\pi}{24}$.

10. 证法：(1) 将不等式左边的每一项改写为二重积分，例如

$$\left(\int_a^b f(x) \cos kx dx \right)^2 = \int_a^b \int_a^b f(x) f(y) \cos kx \cos ky dx dy, \text{ 然后利用三角函数的公式, 又总}$$

有 $f(y) \cos k(x-y) \leq f(y)$, 即可得证。

(2) 将不等式右边与左边的差表示为二重积分，进而证明其被积函数非负。